

Hayatullah Ahmadzai

Durham, North Carolina, USA

ahmadzaihayat@outlook.com | +1 919-408-6341 | Website | Google Scholar | LinkedIn

Education

2015–2019: Ph.D. in Economics (focus: Agricultural and Development Economics), University of Nottingham, UK

2008–2011: M.S. Agricultural Economics, Purdue University, USA

2002–2005: B.S. Agronomy and Soil Sciences, Kabul University, Afghanistan

Primary Research Interests: Agricultural & Food Policy • Climate Change, Adaptation, & Agricultural Resilience • Agribusiness & Farm Economics • Food Security, Rural Livelihoods & Agrifood Systems • Applied Econometrics & Impact Evaluation

Academic and Research Appointments

06/2022–06/2025: Research Fellow, Center for Research in Economic Development and International Trade (CREDIT), University of Nottingham, Nottingham, UK.

04/2019–04/2022: Postdoctoral Research Fellow, International Center for Biosaline Agriculture, Dubai, UAE.

02/2012–09/2015: Senior Economist/Head of Monitoring, Evaluation & Learning (MEL), Department for International Development (DfID, now FCDO), Kabul, Afghanistan.

03/2011–09/2015: Assistant Professor (Teaching-focused, Part-time), Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan.

07/2011–02/2012: Economist and Value-Chain Development Specialist, FLAG International Afghanistan (FIA), Kabul, Afghanistan.

03/2006–10/2007: Teaching Assistant, Department of Agronomy and Soil Sciences, Kabul University, Kabul, Afghanistan.

Fellowships and Scholarly Appointments

2025–2027: MESA Global Academy Scholar, Middle East Studies Association (MESA), Washington, DC, USA (remote, based in Durham, NC, USA).

Awarded a one-year scholarship (2025–2026), competitively renewed for a second year (2026–2027). Conducting policy-relevant research and engaging with scholars on Middle East and North Africa issues.

06/2025–Present: External Research Fellow, Center for Research in Economic Development and International Trade (CREDIT), University of Nottingham, Nottingham, UK (remote, based in Durham, NC, USA). Continuing collaborative research with CREDIT on gender, conflict, and rural household welfare in Afghanistan, leading a working paper on security, infrastructure, and the gender gap in rural labor markets using repeated cross-sectional survey data

Publications

Peer-reviewed Journal Articles

Climate shocks, household food security and welfare in Afghanistan

Ahmadzai, H., and Morrissey, O. 2025. *Food Policy*, 134, 102910.

Crop Diversification and Farm Household Welfare during Conflict: Afghanistan 2011–2017

Ahmadzai, H., and Morrissey, O. 2025. *The Journal of Development Studies*, 61(3), 381–399.

Tests for market failures and overcoming transaction costs barriers to market participation in Afghanistan

Ahmadzai, H. 2024. *Journal of Social and Economic Development*.

Empirical Analysis of Impacts of Sand and Dust Storms on Agricultural Productivity in Mongolia

Ahmadzai, H., Malhotra, A., and Tutundjian, S. 2023. *PLOS ONE*, 18(2), e0269271.

The Impact of Sand and Dust Storms on Agriculture Productivity in Iraq

Ahmadzai, H. 2023. *Middle Eastern Development Journal*, 15(1), 50–65.

Policies for Sustainable Agriculture and Livelihood in Marginal Lands: A Review

Ahmadzai, H., Elouafi, I., and Tutundjian, S. 2022. *Sustainability*, 13(16), 8692.

Orphan Crops: A Best Fit for Dietary Enrichment and Diversification in Highly Deteriorated Marginal Environments

Talabi, A., Vikram, P., Thushar, S., Hifzurrahman, R., Ahmadzai, H., Nhamo, N., Shahid, M., and Singh, R. 2022. *Frontiers in Plant Science*, 13, 839704.

Hope for Change: Is Diversifying Production Portfolios an Ideal Strategy to Boost Farming Efficiency in Afghanistan?

Ahmadzai, H. 2021. *Progress in Development Studies*, 21(3), 1–25.

Development of Quinoa Value Chain to Improve Food and Nutritional Security in Rural Communities in Rehamna, Morocco: Lessons Learned and Perspectives

Hirich, A., Rafik, S., Rahmani, M., Fetouab, A., Filali, K., Ahmadzai, H., et al. 2021. *Plants*, 10(2), 301.

How is Off-farm Income Linked to On-farm Diversification? Evidence from Afghanistan

Ahmadzai, H. 2020. *Studies in Agricultural Economics*, 122(1), 1–12.

Trends in Quinoa Adoption in Marginal Areas, Economic Viability, and Policy Outlook

Ahmadzai, H. 2020. *Journal of Agribusiness and Rural Development*, 57(3), 235–247.

Book Chapters

Marginal Areas and Indigenous People Priorities for Research and Action

Sayed, A., and Ahmadzai, H., et al. 2023. In *Science and Innovations for Food Systems Transformation*. Springer.

Impacts of Sand and Dust Storms on Agricultural Production

Middleton, N., Ahmadzai, H., and Ayurzana, E. 2023. In *Sand and Dust Storms: A Guide to Mitigation, Adaptation, Policy and Risk Management Measures in Agriculture*.

From Desert Farm to Fork: Value Chain Development for Innovative Salicornia-based Food Products—A Case Study in the United Arab Emirates

Lyra, D., Lampakis, E., et al., Ahmadzai, H., et al. 2021. In *Saline Futures: State-of-the-Art in Saline Agriculture*.

Policy Papers & Technical Reports

Marginal Lands: Potential for Agricultural Development, Food Security and Poverty Reduction

Ahmadzai, H., Tutundjian, S., et al. 2022. *SOLAW21 Technical Background Report*, FAO.

Agri-tech: A Catalyst for Treated Wastewater Deployment in High Efficiency Food Production Systems

Ammar, K., Fatnassi, H., and Ahmadzai, H., et al. 2020. *International Center for Biosaline Agriculture (ICBA)*.

Marginal Areas and Indigenous People: Priorities for Research and Action

Sayed, A., and Ahmadzai, H., et al. 2022. *Food Systems Summit Brief*.

Addressing Sand and Dust Storms in Northeast Asia (China and Mongolia) and Western Asia (Iran, Iraq, and Kuwait) and North Africa (Algeria): An Agricultural Perspective

Ahmadzai, H. 2022. *FAO and ICBA*.

Baseline Survey Report: Improving Agricultural Resilience to Salinity (RESADE)

Ahmadzai, H. 2020. International Center for Biosaline Agriculture (ICBA), IFAD, BADEA.

Working Papers

Security, Infrastructure, and Gender Gap in Rural Labor Markets: Evidence from Afghanistan's Repeated Cross-Sectional Survey

Ahmadzai, H.

CREDIT Research Paper; work in progress

Research Grants and Projects

2022–2024: Institute of International Education (IIE) and CARA (Council for At-Risk Academics), IIE-SRF Afghanistan Emergency Initiative Fund. **Principal Investigator.** £86,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Catalyzing Investments and Actions to Enhance Resilience Against Sand and Dust Storms in Agriculture. **Principal Investigator.** \$150,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Climate-Resilient and Gender-Inclusive Cropping Systems in Climate-Vulnerable Areas in Asia and the Pacific. **Co-Principal Investigator** with R.K. Singh (ICBA). \$178,441.

2021: Ministry of Climate Change and Environment, UAE, Developing National Policy to Utilize Treated Sewage Effluent (TSE) and Biosolids in Controlled Environment Agriculture in the UAE. **Co-Principal Investigator** with Khalil Ammar (ICBA). \$100,000.

2020: International Center for Biosaline Agriculture (ICBA), Adoption and Impact of Quinoa Among Farmers in Marginal Areas of the MENA Region. **Principal Investigator.** \$68,488.

2020: University of Wollongong in Dubai and ICBA, Resilient Agrifood Dynamism Through Evidence-Based Policies (READY). **Co-Principal Investigator** with Ioannis Manikas (University of Wollongong) and Khalil Ammar (ICBA). \$1,500,000.

2019: IFAD and BADEA, Improving Agricultural Resilience to Salinity and Climate Change through Development of Pro-Poor Technologies and Management Strategies (RESADE). **Co-Principal Investigator** with R.K. Singh (ICBA). \$7,700,000.

2019: International Center for Biosaline Agriculture (ICBA), EXPO Live Phase II: Value Chain Development for Innovative Halophyte-Based Food Systems. **Co-Principal Investigator** with Dionysia Angeliki Lyra (ICBA). \$500,000.

Courses Taught

Assistant Professor (Part-time), Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan.

2011–2015: AgEcEx731: Agricultural Production Economics (undergraduate, spring terms)

2011–2015: AgEcEx836: Agricultural Finance (undergraduate, fall terms)

2012–2014: AgEcEx630: Introductory Statistics & Econometrics (undergraduate, spring terms)

2011–2015: AgEcEx735: Research Methods & Projects (undergraduate; seminar-based teaching and student research supervision)

Selected Presentations

Conference Presentations

2026: “Climate Shocks, Household Food Security, and Welfare in Afghanistan.” Stanford University (FSE, Center for South Asia, Abbasi Program in Islamic Studies), April 22, 2026. Event | Slides | Video

2023: “Crop Diversification and Farm Household Welfare during Conflict: Afghanistan 2011–2017.” 6th MaDE Workshop on Emerging and Developing Countries, Loughborough Business School (online), June 5–6, 2023. Slides

2021: “Assessing the Impact of Sand and Dust Storms on Agriculture.” Food and Agriculture Organization (FAO), Bangkok (online), April 7–11, 2021. Slides

2020: “Salinization in Marginal Agricultural Areas: Implications for Global Food Security and Poverty.” International Center for Biosaline Agriculture (ICBA), Dubai, UAE, June 27–28, 2020. Slides

2018: “Tests for Market Failures and Overcoming Transaction Costs in Afghanistan.” Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, September 8–12, 2018. Slides

2017: “Status, Patterns, and Microeconomic Drivers of the Extent of Diversity in Crop Production: Evidence from Afghanistan.” Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 15–20, 2017. Slides

2016: “Crop Diversification and Technical Efficiency: Evidence from Afghanistan.” Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 12–16, 2016. Slides

Invited Talks

2021: “Challenge and Opportunity of Quinoa for Food and Nutrition Security.” Panellist with Warren T.K. Lee, chaired by Bo Zhou. FAO Regional Office for Asia and the Pacific (FAO-RAP), Bangkok, Thailand, November 7–8.

2021: “The Economics of Quinoa Production: Opportunity for Marginal Farmers.” Panellist with Dr. José Graziano da Silva. Global Forum on Innovations for Marginal Environments (GFIME), Dubai, UAE, November 20–21.

Awards

2021: Co-recipient, Khalifa International Award for Date Palm and Agricultural Innovation (KIADPAI), awarded as Co-PI of the Project: “From Desert Farm to Fork: Value Chain Development for Innovative Halophyte-Based Food Products.”

2015: PhD Scholarship Award, University of Nottingham and Islamic Development Bank (IsDB).

2007: Graduate Scholarship Award, Purdue University and USAID-funded Afghan Merit Scholar Program (AMSP).

Service and Professional Activities

2019–present: External Research Fellow, Centre for Research in Economic Development and International Trade (CREDIT), University of Nottingham, UK.

2019–present: Member, Agribusiness Sub-network, Association of Agricultural Research Institutions in the Near East and North Africa (AARINENA).

2019–present: Peer Reviewer for *The Journal of Development Studies*; *Food Policy*; *Sustainability* (MDPI); *PLOS ONE*; *Review of Development Economics*; *Humanities and Social Sciences Communications*; *Agriculture* (MDPI); *Cogent Economics and Finance*; *Journal of Social and Economic Development*.

Media and Outreach

2022: “How COVID-19 Crisis Undermined MENA States’ Food Security.” *Arab News*, November 20, 2022.

2021: “Agriculture in Marginal Lands.” *Scholarly Community Encyclopedia*, September 2021.

2021: “New Study Reviews Status of Marginal Lands Worldwide.” ICBA News Article, August 29, 2021.

2021: “World’s Indigenous Peoples to Bear Brunt of Climate Change, Warns UN Paper.” ICBA News Article, April 25, 2021.

2020: “ICBA’s Quinoa Varieties Show Promise in Morocco.” ICBA News Article, April 15, 2020.

Professional Development

2020: Responsible Conduct in Research: Research, Ethics, and Compliance. Collaborative Institutional Training Initiative (CITI). Completed online, March 24–April 4.

2013: Leadership and Governance of Development Programs and Projects. Crown Agents International. Dubai, United Arab Emirates, November 11–15.

2012: Management of Agriculture and Rural Development Projects. Asian Institute of Technology (AIT). Bangkok, Thailand, November 21–December 1.

2008: Academic English and Cross-Cultural Training Program. International Centre for Intercultural Communication (ICIC). IUPUI, Indianapolis, IN, USA, October 29, 2007–May 30, 2008.

Skills

- **Economic & Policy Analysis:** Impact evaluation, program evaluation, benefit-cost analysis, value-for-money analysis, cost-effectiveness analysis, project appraisal, feasibility studies, policy analysis and design, welfare and distributional analysis, economic valuation, decision analysis.
- **Quantitative Methods:** Applied econometrics, causal inference, quasi-experimental methods (IV, DiD, RDD, matching, discrete choice, panel data), experimental methods (RCTs, field experiments), survey design and sampling, household survey design and implementation, mixed-methods research.
- **Project and Grant Management:** Grant writing and proposal development, multi-partner project management, interdisciplinary collaboration, stakeholder engagement, capacity building and mentoring, technical and policy reporting, scientific writing, theory of change development, M&E framework design.
- **Software and Tools:** Stata, R, Python (Pandas, NumPy, Statsmodels, Seaborn, Matplotlib), Power BI, QGIS, Excel, Survey Solutions, Qualtrics, LaTeX, Markdown.

Linguistic

- English (fluent)
- Pashto (native)
- Dari/Persian (fluent)
- Arabic (conversational)